LAB EXPERIMENT RECORD

13-10-2026---Set 18: Product Inventory Management

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	13 OCTOBER 2026
Day	TUESDAY

Aim

To analyze product inventory data using a bar chart, stacked bar chart and table, and to build an interactive Tableau dashboard for exploring the inventory data.

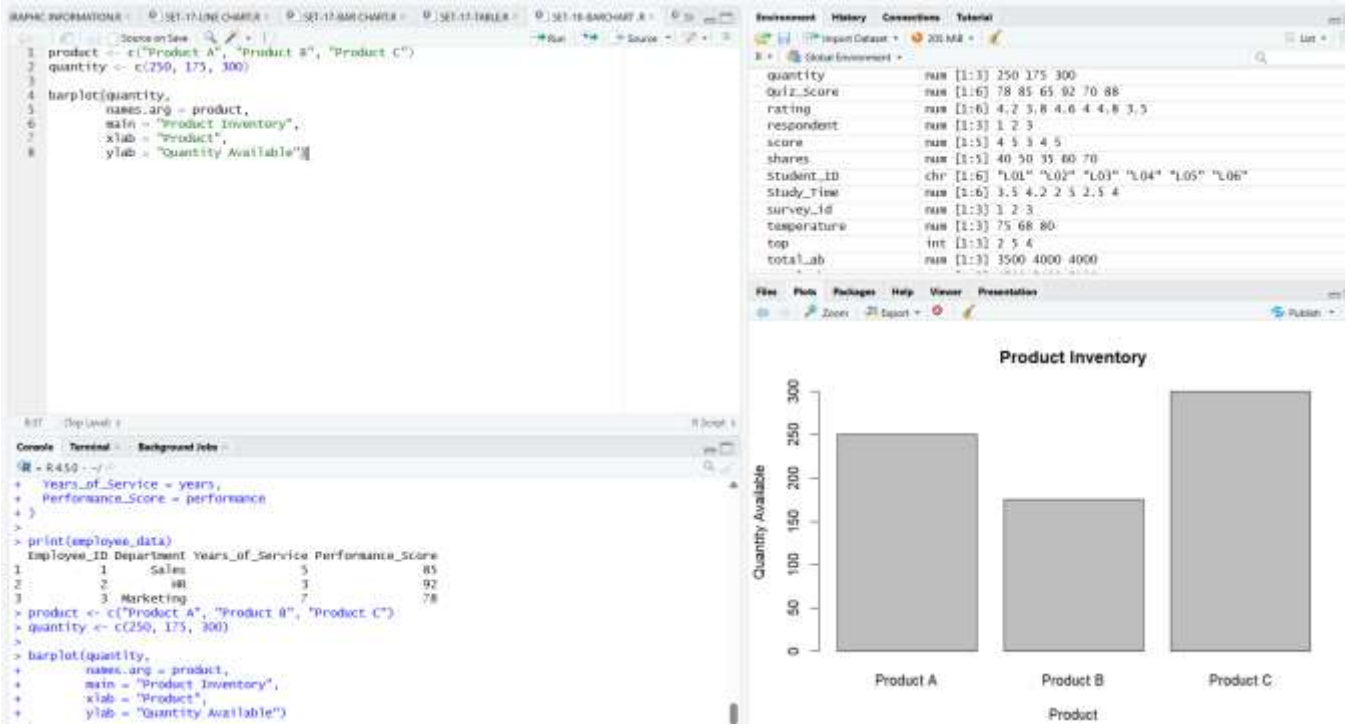
Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Question 1: Bar Chart of Product Quantity

Question: Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Aim: To visualize the quantity available for each product using a bar chart with labelled axes and a title.

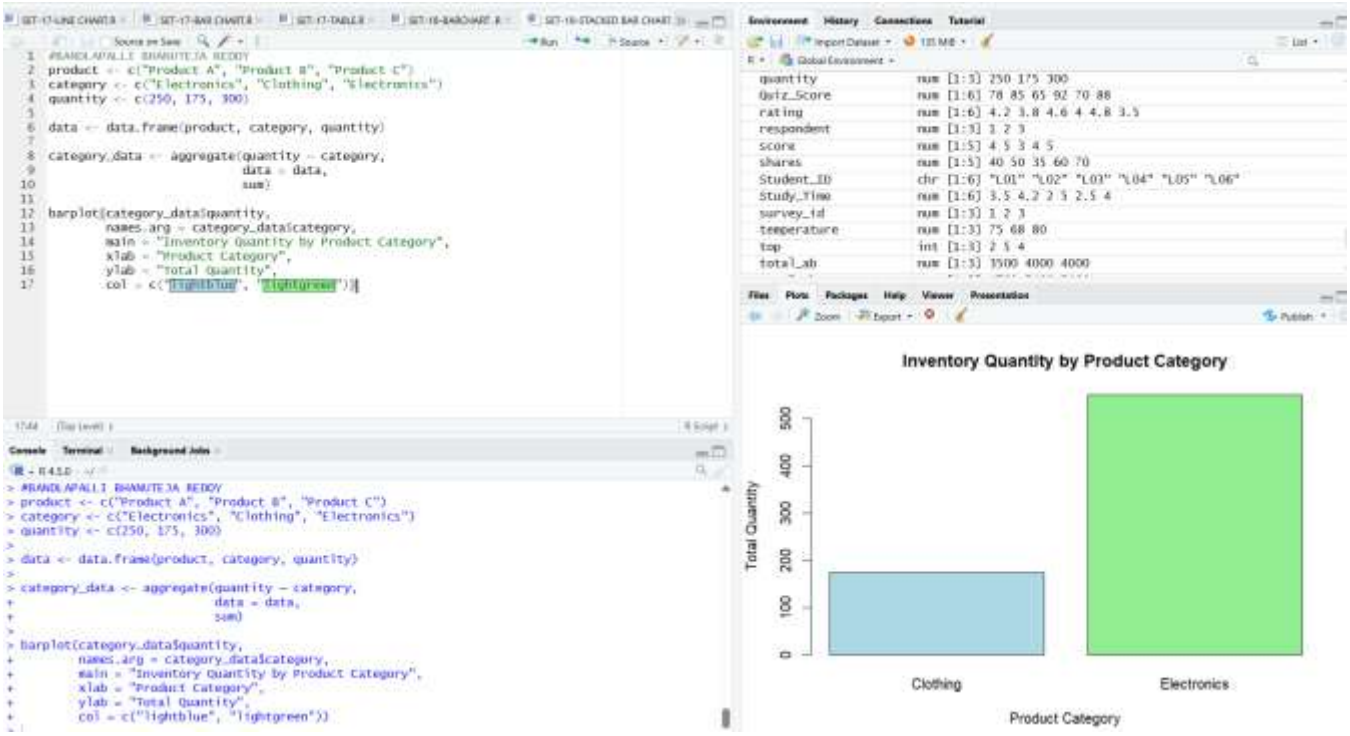


Result: Product C has the highest stock (300) and Product B the lowest (175), so Product B needs restocking first.

Question 2: Stacked Bar Chart by Product Category

Question: Generate a stacked bar chart to show the quantity of each product within different product categories.

Aim: To show how the quantity of each product is divided across product categories using a stacked bar chart.

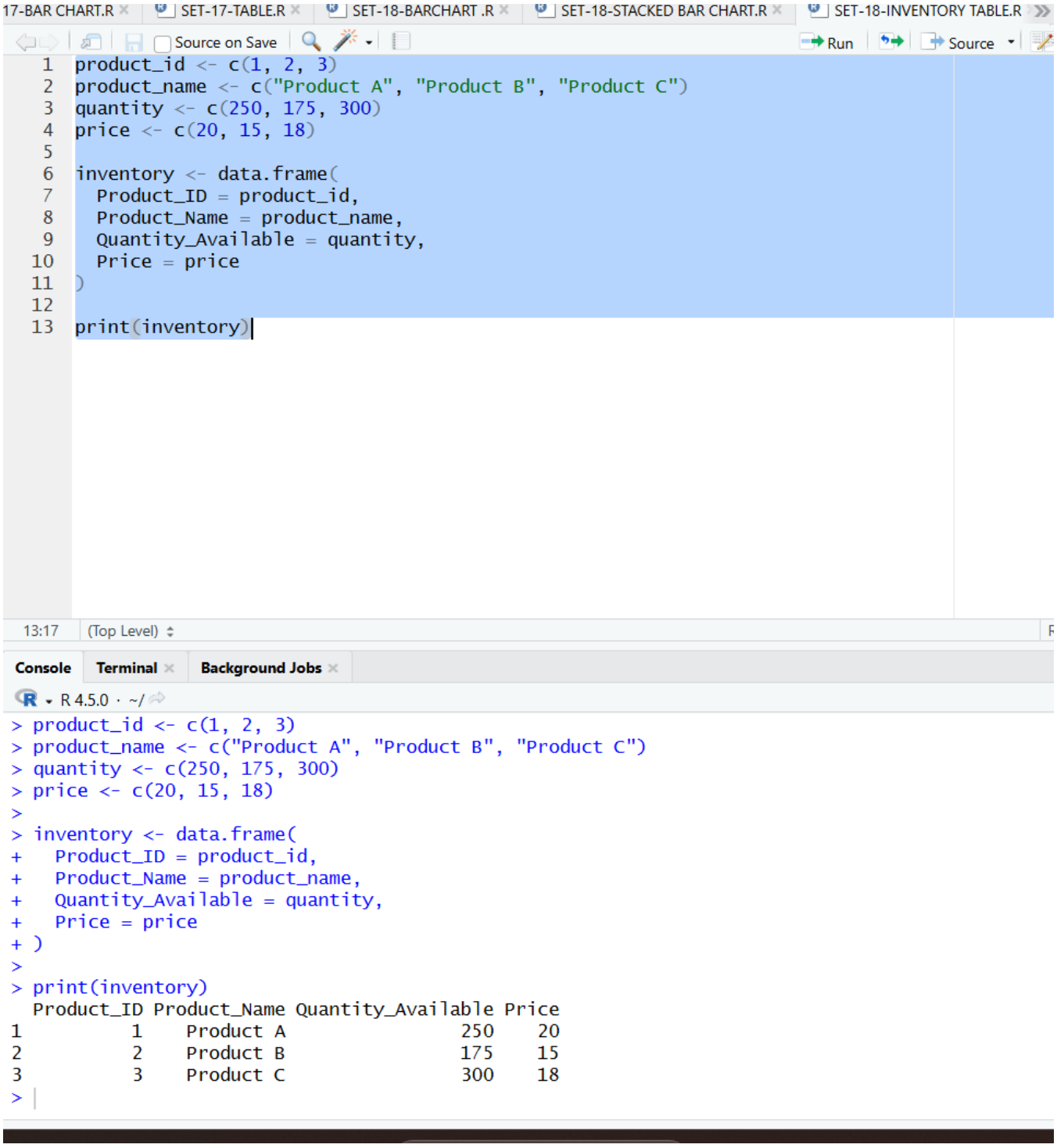


Result: Each bar is split by category, showing each category's share of a product's stock; the dataset gives no category column, so categories must be supplied.

Question 3: Table of Inventory Data

Question: Build a table to show the inventory data for each product. Label the table elements.

Aim: To present the quantity and price of each product in a clearly labelled table.



Result: The table lists all three products; the stock values are 5000 for Product A, 2625 for Product B and 5400 for Product C, so Product C holds the most value.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Aim: To build an interactive Tableau dashboard that combines the bar chart, stacked bar chart and table for exploring inventory data.



Result: The dashboard places all three views together, and a product filter updates them so users can explore the inventory interactively.

Overall Result

Observation: Total stock is 725 units with a total value of 13,025, and Product C is the largest item in both quantity (300) and value (5400).